

4.7.4 The rate and extent of chemical change

Chemical reactions can occur at very different rates. Although the reactivity of chemicals is a significant factor in how fast chemical reactions proceed, there are many variables that can be manipulated in order to speed them up or slow them down. Catalysts, including enzymes, can have a very significant effect on reaction rates. An understanding of the energy changes that accompany bond breaking and bond forming can help to account for the effect of temperature on rates. Chemical reactions may also be reversible and therefore the effect of different variables needs to be established in order to identify how to maximise the yield of desired product.

There are two required practicals: an investigation of the effect of changes in concentration on the rate of a chemical reaction and an investigation of a factor affecting the rate of an enzyme-controlled reaction.

4.7.4.1 Factors that affect reaction rates

GCSE Science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe the effect of changes in temperature, concentration, pressure, and surface area on rate of reaction. Suggest practical methods for determining the rate of a given reaction.	The rate of a chemical reaction can be found by measuring the quantity of a reactant used or the quantity of product formed over time: mean rate of reaction = $\frac{\text{quantity of reactant used}}{\text{time taken}}$ mean rate of reaction = $\frac{\text{quantity of product formed}}{\text{time taken}}$ The quantity of reactant or product can be measured by the mass in grams, by a volume in cm³ ((HT only) or by an amount in moles). The units of rate of reaction may be given as g/s, cm³/s ((HT only) or mol/s). The rate of a chemical reaction can be determined by measuring: • the loss in mass of a reactant's mixture • the volume of gas produced • the time for a solution to become opaque or coloured.	WS 3.3, MS 1a, 1c Calculate the mean rate of a reaction from given information about the quantity of a reactant used or the quantity of a product formed and the time taken. WS 3.5, MS 4a, 4b, 4c Draw, and interpret, graphs showing the quantity of product formed or quantity of reactant used up against time to compare or determine rates of reaction. WS 3.3, MS 4e Draw tangents to the curves on these graphs and use the gradient of the tangent as a measure of the rate of reaction. WS 3.3, MS 4d, 4e ((HT only) Calculate the gradient of a tangent to the curve on these graphs as a measure of rate of reaction at a specific time.

4.7.4.2 The effect of surface area on rates of reaction

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain the effects on rates of reaction of changes in the size of the pieces of a reacting solid in terms of surface area to volume ratio.	Breaking up a solid reactant into smaller pieces increases the surface area that can be in contact with any solution with which it reacts. Increasing the ratio of surface area to volume increases the rate of reaction for a given mass of a solid reactant.	MS 1c Use proportionality when comparing factors affecting rate of reaction. MS 5c Calculate surface areas and volumes of cubes.

4.7.4.3 The effect of temperature, concentration and pressure on rates of reaction

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain the effects on rates of reaction of changes in temperature, concentration and pressure in terms of the frequency and energy of collision between particles.	According to collision theory, chemical reactions can occur only when reacting particles collide with each other and with sufficient energy. The minimum amount of energy that particles must have to react is called the activation energy. Increasing the concentration of reactants in solution, the pressure of reacting gases and the surface area of solid reactants increases the frequency of collisions and so increases the rate of reaction. Increasing the temperature increases the frequency of collisions and makes the collisions more energetic, and so increases the rate of reaction.	WS 1.2 Predict and explain the effects of changing conditions on the rate of a reaction.

Required practical activity 19: investigation of how changes in concentration affect the rates of reactions by a method involving measuring the volume of a gas produced and a method involving a change in colour or turbidity. This should be an investigation involving developing a hypothesis.

AT skills covered by this practical activity: chemistry AT 1, 3, 5 and 6.

This practical activity also provides opportunities to develop WS and MS. Details of all skills are given in [Key opportunities for skills development](#) (page 176).

4.7.4.4 Activation energy

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain activation energy as the energy needed for a reaction to occur. Draw and label a reaction profile for an exothermic and an endothermic reaction, identifying activation energy.	Chemical reactions can occur only when reacting particles collide with each other and with sufficient energy. The minimum amount of energy that particles must have to react is called the activation energy. Reaction profiles can be used to show the relative energies of reactants and products, the activation energy and the overall energy change of a reaction.	WS 3.2, 3.5, MS 4a Interpret reaction profiles, including using them to identify reactions as exothermic or endothermic.

4.7.4.5 Bond breaking and bond forming (HT only)

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Calculate energy changes in a chemical reaction by considering bond making and bond breaking energies.	During a chemical reaction: - energy must be supplied to break bonds in the reactants - energy is given out when bonds in the products are formed. The energy needed to break bonds and the energy given out when bonds are formed can be calculated from bond energies. The difference between the sum of the energy needed to break bonds in the reactants and the sum of the energy given out when bonds in the products are formed is the overall energy change of the reaction. In an exothermic reaction, the energy given out from forming new bonds is greater than the energy needed to break existing bonds. In an endothermic reaction, the energy needed to break existing bonds is greater than the energy given out from forming new bonds.	WS 3.3, MS 1a Use arithmetic computation when calculating energy changes. WS 1.2, 3.3, MS 1a, 4a Calculate the energy transferred in chemical reactions between simple molecules in the gas state using bond energies supplied.

4.7.4.6 Catalysts

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe the characteristics of catalysts and their effect on rates of reaction. Identify catalysts in reactions. Explain catalytic action in terms of activation energy.	Catalysts change the rate of chemical reactions but are not used up during the reaction. Different reactions need different catalysts. Knowledge of the names of catalysts other than those specified in the subject content is not required.	WS 3.5 Identify catalysts in reactions from their effect on the rate of reaction and because they are not included in the chemical equation for the reaction. WS 1.2 Use reaction profiles to explain catalytic action.

4.7.4.7 Enzymes

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Recall that enzymes act as catalysts in biological systems. Explain the mechanism of enzyme action including the active site, enzyme specificity and factors affecting the rate of enzymatic reaction.	Enzymes are important as biological catalysts which allow all the reactions in cells to occur. Enzymes are large protein molecules. The shape of an enzyme is vital for its function. Each enzyme has an active site with a unique shape to bind a specific substrate molecule. High temperatures and extremes of pH denature the enzyme, changing the shape of the active site. The 'lock and key' model is a simplified model of enzyme action. Different enzymes work fastest at different temperatures and pH values.	WS 3.3, 3.5, MS 1a, 1c, 1d Carry out rate calculations for chemical reactions and make estimates of simple calculations without using a calculator.

Required practical activity 20: investigate the effect of pH on the rate of reaction of amylase enzyme. Students should use a continuous sampling technique to determine the time taken to completely digest a starch solution at a range of pH values. Iodine reagent is to be used to test for starch every 30 seconds. Temperature must be controlled by use of a water bath or electric heater.

AT skills covered by this practical activity: biology AT 1, 2 and 5.

This practical activity also provides opportunities to develop WS and MS. Details of all skills are given in [Key opportunities for skills development](#) (page 177).

4.7.4.8 Reversible reactions

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Recall that some reactions may be reversed by altering the reaction conditions.	In some chemical reactions, the products of the reaction can react to produce the original reactants. Such reactions are called reversible reactions and are represented: $A + B \rightleftharpoons C + D$ The direction of reversible reactions can be changed by changing the temperature. Examples include the effect of changing the temperature on the decomposition of ammonium chloride and of hydrated copper(II) sulfate.	

4.7.4.9 Dynamic equilibrium

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Recall that dynamic equilibrium occurs when the rates of forward and reverse reactions are equal.	When a reversible reaction occurs in apparatus which prevents the escape of reactants and products, equilibrium is reached when the forward and reverse reactions occur at exactly the same rate.	

4.7.4.10 Factors affecting the position of equilibrium (HT only)

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Predict the effect of changing reaction conditions on equilibrium position.	The relative amounts of all the reactants and products at equilibrium depend on the conditions of the reaction. If a system is at equilibrium and a change is made to any of the conditions, then the system responds to counteract the change.	WS 1.2 Apply Le Châtelier's principle to make qualitative predictions about the effect of changes on systems at equilibrium when given appropriate information.

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Predict the effect of changing concentration on equilibrium position and suggest appropriate conditions to produce a particular product.	If the concentration of one of the reactants or products is changed, the system is no longer at equilibrium and the concentrations of all the substances change until equilibrium is reached again. If the concentration of a reactant is increased, more products form until equilibrium is reached again. If the concentration of a product is decreased, more reactants react until equilibrium is reached again.	WS 3.5 Interpret appropriate given data to predict the effect of a change in concentration of a reactant or product on given reactions at equilibrium.
Predict the effect of changing temperature on equilibrium position and suggest appropriate conditions to produce a particular product.	If the temperature of a system at equilibrium is increased: - the relative amount of products at equilibrium increases for an endothermic reaction - the relative amount of products at equilibrium decreases for an exothermic reaction. If the temperature of a system at equilibrium is decreased: - the relative amount of products at equilibrium decreases for an endothermic reaction - the relative amount of products at equilibrium increases for an exothermic reaction.	WS 1.2 Apply the idea that if a reversible reaction is exothermic in one direction, it is endothermic in the opposite direction.
Predict the effect of changing pressure on equilibrium position and suggest appropriate conditions to produce a particular product.	For gaseous reactions at equilibrium: - an increase in pressure causes the equilibrium position to shift towards the side with the smaller number of molecules, as shown by the symbol equation for that reaction - a decrease in pressure causes the equilibrium position to shift towards the side with the larger number of molecules, as shown by the symbol equation for that reaction.	